Postpartum Hemorrhage

Definition:

- Post partum hemorrhage is excessive blood loss from the genital tract at any time following the baby"s birth up to 6 weeks, the blood loss is ufficient to affect the general condition of the mother as shown by tachycardia and / or hypotention and poor peripheral perfusion.
- The traditional definition based on a blood loss greater than 500 mL during a vaginal delivery or greater than 1,000 mL with a cesarean delivery after delivery, is difficult to estimat in clinical practice.

Incidence:

- In Egypt, postpartum hemorrhage is the attributed cause for 32% of all maternal deaths, and 46% of all direct maternal death.
- 20-60% of maternal deaths each year in the world are due to PPH
- 14 million cases of PPH per year worldwide

Types of post partum hemorrhage (PPH):

1 – **Primary PPH** ⟶ occurs during the third stage of labor or within 24 hours of delivery

2 – **Secondary PPH** ⟶ occurs after the first 24 hours up to the 42nd day

Predisposing factors:

- Previous history of postpartum haemorrhage or retained placenta
- High parity
- Presence of fibroids
- Maternal anaemia
- Ketoacidosis
- First pregnancy
- Large baby
- Multiple pregnancy
- Hydramnios
- Antepartum hemorrhage
- Preeclampsia
- Augmented labor

- Rapid labor
- Prolonged third stage of labor
- Episiotomy
- Operative delivery
- Chorioamnonitis
- Use of uterine relaxing agents

Causes: (The "Four Ts")

1 –**Tone 70%**

Uterine atony → poor or ineffective uterine contractios and retraction

2 –**Trauma 20%**

Genital tract truma → Lacerations, hematomas, inversion, rupture

3 – Tissue 10% (**Third stage complications)**:

- Missmanagment of third stage
- Acute inversion of the uterus
- Abnormal or incomplete placental separation
- Retained tissue and placenta

4 – Thrombin → Coagulation disorders

1 – **Factors predisposed to uterine atony:**

- Overdistended uterus due to polyhydamnios, multiple pregnancy , or macrosomia
- Uterine muscle exhaustion due to precipitate labor , prolonged labor or grand multiparity
- Amniotic infection due to prolonged ROM and Chorioamnonitis
- Functional or anatomic distortion of the uterus due to a fibroid uterus, placenta previa
- General anesthetics (e.g., halothane)
- Incomplete separation of the placenta

2 - **Factors predisposed to genital tract truma**

- A precipitate delivery predisposes the genital tract lacerations of the cervix , vagina or perineum
- Malposition or deep engagement of the fetal head
- Previous uterine surgery → uterine rupture

3 - **Factors predisposed to retained products of conception or retained placenta**

- An incomplete placenta at delivery
- Previous uterine surgery
- High parity

- An abnormal placenta on ultrasound

4 – Abnormalities of coagulation

- Hemophilia A
- History of liver disease
- Use of anticoagulants

Signs & symptoms of postpartum hemorrhage

a) If the bleeding due to uterine atony: (Primary PPH)

- ↑ amount of blood loss
- Shock → ↓ BP, ↑ pulse, thirst, dyspnea and pallor
- Uterus not contracted, increase in size and soft.
- Color of the blood→ dark venous
- Fundal level is rising

b) If the bleeding due to laceration (Primary PPH)

- ↑ amount of blood loss
- Uterus is contracted, firm and normal in size
- Bleeding is appear from lacerated site
- The blood is red in color

c) In case of secondary PPH:

- Bleeding variable (light or heavy, continuous or irregular)
- Uterus is softer and increase in size
- Foul – smelling - Anemia

Diagnosis

A - History

- Personal history including name and age
- Medical history including history of hypertension ,diabetes , anemia
- Surgical history including previous uterine surgery
- History of previous blood transfustion
- Obstetric history including :
- History of any of the previously mentioned risk factors
- Number, mode and outcome of previous deliveries
- Obstructed labor or insterumental

B - Physical examination

1 – General examination:

- Vital signs → Tachycardia and hypotension
- Shock manifested early by dizziness, weakness or sweating or manifested late by restlessness, pallor, shortness of breath and /or collapse.

2 – Abdominal examination :

- Consistency of the uterus (lax or firm)
- Uterine fundal level
- Any abdominal tenderness or rigidity
- Scars from previous operations

3 – Local examination :

- Assess the amount of vaginal bleeding
- Look for lacerations (perineal , vulvar , vaginal or cervical)
- Determine whether the placenta has been delivered

4 – Laboratory investigation :

- ABO grouping and Rh type
- CBC (Hb ,Hct ,platelet count)
- Bleeding time
- Coagulation time

Diagnosis of PPH

Presenting symptoms and other symptoms and signs typically present	Symptoms and signs sometimes present	Probable diagnosis
- Primary PPH - Uterus soft & not contracted	Shock	Atonic uterus
- Primary PPH	-Complete placenta -Uterus contracted	Tears of cervix , vagina or perineum
Placenta not delivered within 30 minutes after delivery	- Primary PPH -Uterus contracted	Retained placenta
Portion of the maternal surface of the placenta missing or torn membranes with vessels	-Immediate PPH - Uterus contracted	Retained placental fragments
- Uterine fundus not felt on abdominal palpation - Slight or intense pain	- Inverted uterus apparent at vulva - Primary PPH	Inverted uterus
PPH (bleeding is intra-abdominal pain and /or vaginal) -Sever abdominal pain	- Shock - Tender abdomen - Rapid maternal pulse	Ruptuered uterus
-Bleeding occurs more than 24 h after delivery - Uterus ,softer and larger than expected for elapsed time since delivery	- Bleeding is variable (light or heavy , continous or irregular) foul – smelling Anemia	Secondary PPH

Management

1 - Prophylactic measures:

a) Preventive care during the antenatal period:

- Recognize risk factors by taking complete history
- Perform routine laboratory investigation to diagnose anemia at an early stage
- Provide health education
- Ensure iron folic acid supplementation
- Educate the patient regarding the importance of the following:
    - Anemia prevention

- Going to the hospital early in labor
- Seeking immediate medical attention if vaginal bleeding occurs

b) Preventive care during Labor:

- Routine laboratory investigations. If the hemoglobin level is < 8 gm /dl crossmatch at least one unit of whole blood even the delivery is normal, or two units if it is CS.

- Identify the patients at risk of developing PPH

- Avoid bearing down in the first stage of labor

- Proper using of sedative and anesthesia

- Prophylactic administration of oxytocic drugs with delivery of anterior shoulder or at the end of third stage.

- **Actively manage the third stage of labor as**

  - Administer 10 IU of oxytocine IM with delivery of the anterior shoulder.
  - Perform controlled cord traction to deliver the placenta
  - No attempts for removal of the placenta befor separation
  - Evacuation of the bladder and rectum
  - Avoid massaging the uterus before separation of the placenta
  - Examine the placenta after delivery
  - IV ergometrine after delivery of the placenta
  - Examine the utero-vaginal canal for trauma and prompt repair if present.
  - Effective management of the fourth stage.

Active management

Aime:

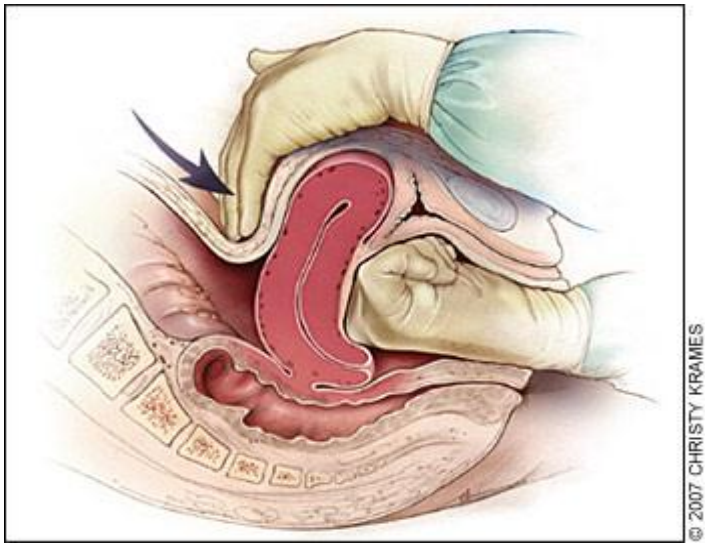
1 – Stop the bleeding

2 – Restore the blood loss and resuscitate the mother

In case of uterine atony

1 - Exploration of uterus under general anesthetic

2 - Bimanual compression of the uterus, massage is performed by placing one hand in the vagina and pushing against the body of the uterus while the other hand compresses the fundus from above through the abdominal wall (Uterus is firmly squeezed between 2 hands). The posterior aspect of the uterus is massaged with the abdominal hand and the anterior aspect with the vaginal hand.



Bimanual uterine compression massage

3 - Uterotonic Agents

a- **Oxytocin**:

10 IU: IM or IV: infuse 20 IU in 1 L of saline at a rate of 250 mL/h (60 drops /m). Maximum dose: 3L of IV fluids containing oxytocin

b – Methergine:

IM or IV (slowly: 0.2 mg If requiered, give 0. 2 mg IM or IV slowly / 4 h**.** Maximum dose: 5 doses (Total 1.0 mg)

c – Prostaglandins:

IM in a dose of 0.25 mg; can be repeated every 15 minutes for a total dose of 2 mg.

4 - Tight intrauterine packing to exert direct hemostatic pressure on the open uterine sinuses and to stimulate uterine contractions.

5 - If all the above measures fail to achieve hemostasis a hysterectomy is performed.

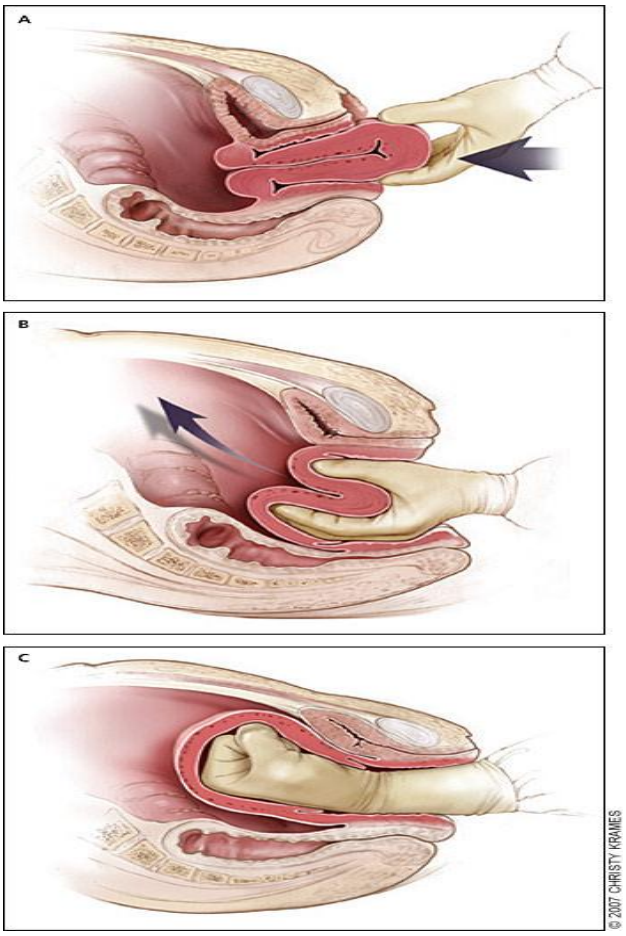
TRAUMA

- Speculum examination done to find out trauma and hemostasis is achieved by appropriate sutures if direct pressure does not stop the bleeding.
- Repair of cervical , vaginal perineum tear
- **A laporotomy** may be required if the cervical tear has extend deep beyond the vaginal valut

Uterine Inversion

- The Johnson method of reduction begins with grasping the protruding fundus with the palm of the hand and fingers directed toward the posterior fornix. The uterus is returned to position by lifting it up through the pelvis and into the abdomen.
- Once the uterus is reverted, uterotonic agents should be given to promote uterine tone and to prevent recurrence.

The Johnson method



Retained placenta

- Determine the cause of retained placenta
- Administer ergotrate 0.2 mg (1 ampoule) IM or misoprostol 200ug (one tablet) rectally followed by controlled cord traction to deliver the placenta.
- If the bleeding persists without delivery of the placenta
- Manually remove the placenta with adequate sedation or anesthesia

In rupture of the uterus

It is treated by either → repair or abdominal hysterectomy

Nursing Management

1 - Assessment

- Identify Risk Factors in the Patient's History
- **Assess:**
  - Vital signs and general condition of the mother.
  - State of uterus.
  - Nature of bleeding.
  - Signs and symptoms of blood loss.
  - Amount of blood loss.
  - Compare laboratory reports.

2 - Nursing Interventions

- Inform the obstetrician.
- Insert two IV canola (size 16 or 18)
- CBC for cross matching and required investigation before starting any IV fluids
- Folly catheter or encourage women to empty the bladder
- Oxygen at mask
- Monitor urine out put
- Examine the placenta and membranes for completeness.
- Provide warmth
- Observe amount of blood loss and vaginal discharge
- Change perineal pad
- Keep all pads and linen to assess the volume of blood lost
- Assess the general physical condition of the mother. (Face, skin...)
- Monitor vital signs / 15minutes.
- Put the infant to the breast to suck or stimulate the nipple manually.

- Prepare instruments and equipment such as sterile gloves, cannula # 18, IV set, catheter set.... etc.
- Start IV infusion and oxytocin drip as ordered.
- Administer medications as ordered.
- Reassure the mother → never leave the mother alone.
- Touch the mother's hand and talk to her

1 - If atonics uterus:

- Feel consistency of the uterus → the fundus is first gently felt with 10

  the fingertips to assess its consistency.
- If it is soft and relaxed the fundus is massaged with a smooth circular motion, applying no undue pressure. When a contraction occurs the hand is held still.
- If the bleeding persists prepare patient for surgery

2 - In cases of traumatic bleeding:

- Press on the tear or laceration.
- Prepear equipment and instruments which required for repair
- Repair tear and laceration